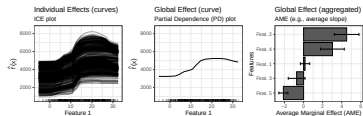
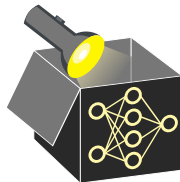


# Interpretable Machine Learning

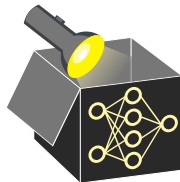
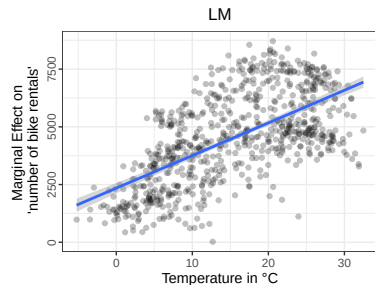
## Feature Effects Introduction



### Learning goals

- Global Feature Effects
- Local Feature Effects

# FEATURE EFFECTS - GLOBAL VIEW



LM without interaction:

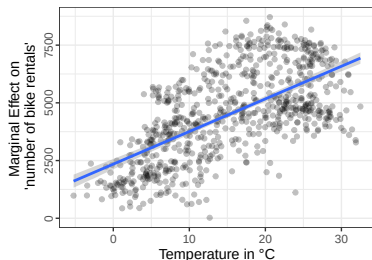
$\hat{\theta}_j$  is linear effect of feature  $x_j$

(applies globally to all observations):

- Model equation:  $\hat{f}(\mathbf{x}) = \hat{\theta}_0 + x_1 \hat{\theta}_1$
- Scalar  $\hat{\theta}_1$  describes global effect

# FEATURE EFFECTS - GLOBAL VIEW

LM

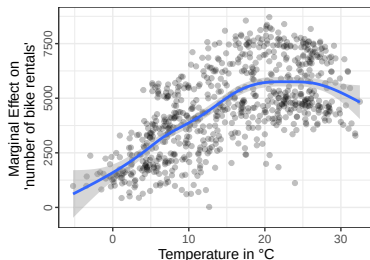


LM without interaction:

$\hat{\theta}_j$  is linear effect of feature  $x_j$   
(applies globally to all observations):

- Model equation:  $\hat{f}(\mathbf{x}) = \hat{\theta}_0 + x_1 \hat{\theta}_1$
- Scalar  $\hat{\theta}_1$  describes global effect

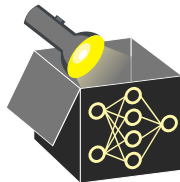
GAM



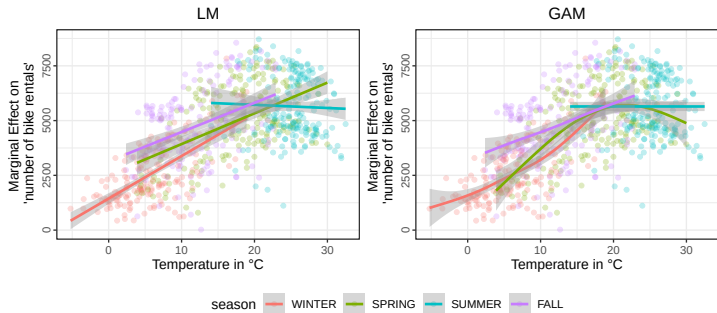
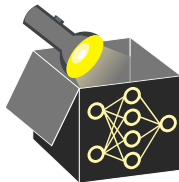
GAM without interaction:

$\hat{f}_j(x_j)$  is non-lin. effect of feature  $x_j$   
(applies globally to all observations):

- Model equation:  $\hat{f}(\mathbf{x}) = \hat{\theta}_0 + \hat{f}_1(x_1)$
- Curve  $\hat{f}_1$  describes global effect



# FEATURE EFFECTS - LOCALIZED VIEW

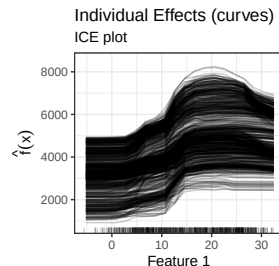
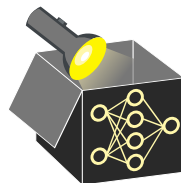


- **Interactions:** Feature effect depends on other features and varies across obs.
  - ⇒ E.g., effect of **temperature** varies across **season**
  - ⇒ Multiple values / curves needed to describe effect
- ML models capture non-linear effects and high-order interactions
  - ⇒ Global view may mislead (single curve may fail to capture complexity)
  - ⇒ Local feat. effect methods needed to estimate effects for individ. obs.
  - ⇒ Global view can be reconstructed by aggregating local effects

# FEATURE EFFECTS

**Feature effects** visualize or quantify how model predictions change as a single feature varies, while all other features are held fixed.

- Analogous to regression coefficients (LMs) or Splines (GAMs)
- Different aggregation levels exist (simplification but information loss)
- Methods: ICE curves (local curves)

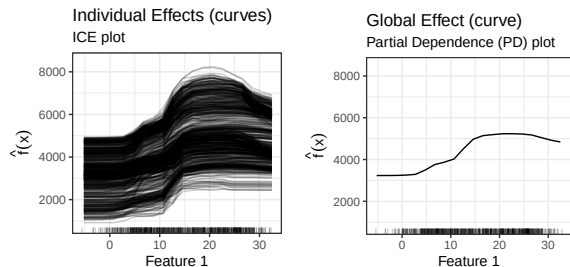
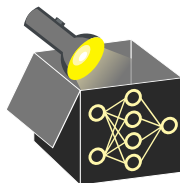


Individual (curves)

# FEATURE EFFECTS

**Feature effects** visualize or quantify how model predictions change as a single feature varies, while all other features are held fixed.

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- Different aggregation levels exist (simplification but information loss)
- Methods: ICE curves (local curves), PD and ALE plots (global curves)

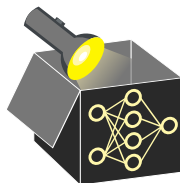


Individual (curves)  $\xrightarrow[\text{curves}]{\text{aggregate}}$  Global (single curve)

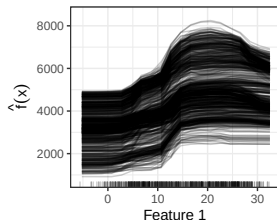
# FEATURE EFFECTS

**Feature effects** visualize or quantify how model predictions change as a single feature varies, while all other features are held fixed.

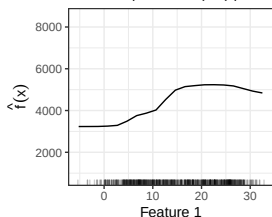
- Analogous to regression coefficients (LMs) or Splines (GAMs)
- Different aggregation levels exist (simplification but information loss)
- Methods: ICE curves (local curves), PD and ALE plots (global curves), AME (global value)



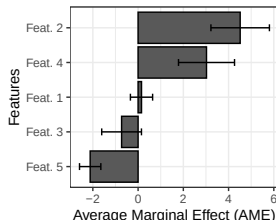
Individual Effects (curves)  
ICE plot



Global Effect (curve)  
Partial Dependence (PD) plot



Global Effect (aggregated)  
AME (e.g., average slope)



Individual (curves)  $\xrightarrow[\text{curves}]{\text{aggregate}}$  Global (single curve)  $\xrightarrow[\text{slopes}]{\text{aggregate}}$  Global (single value)